Yueqing Xuan

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Education

PhD in Computer Science, RMIT University Aug 2022 – Feb 2026

- **Supervisors**: Prof Jeffrey Chan, Prof Mark Sanderson and Dr Kacper Sokol (USI Lugano).
- **Scholarship**: Recipient of a PhD Scholarship funded by the *Australian Research Council (ARC) Centre of Excellence for Automated Decision-Making and Society (ADM+S)*.
- **Research focus**: Explainable AI, Responsible AI, Recommender Systems, Counterfactual Explanations, Human–Computer Interaction.
- **Thesis**: Accounting for User Diversity in the Pursuit of Fair and Explainable Machine Learning.

Master of Information Technology, University of Melbourne Feb 2020 – Jul 2022

- **Specialisation**: Artificial Intelligence.
- **Supervisors**: Dr Lea Frermann and Dr Adi Tarigoppula.
- **Achievements**: WAM 84.50 (H1, First Class Honours); recognised on the *Dean's Honours List 2021* for ranking in the top 5% of students; awarded the *Melbourne Graduate Scholarship 2022*.

Bachelor of Commerce, University of New South Wales Jul 2015 – Jul 2018

- **Major**: Accounting and Information Systems.

Publications

Conference and Journal Papers

[1] **Yueqing Xuan**, Kacper Sokol, Danula Hettiachchi. "Explanation Navigator: Rectifying Out-of-Scope Human Interpretations of Leaky AI Explanations through Conversational Guidance". In: arXiv preprint arXiv:2609.22707. 2026.

[2] **Yueqing Xuan**, Kacper Sokol, Mark Sanderson, Jeffrey Chan. "Diversity-Augmented Negative Sampling for Implicit Collaborative Filtering". Proceedings of the ACM Web Conference 2026 (WWW'26), 2026, https://doi.org/10.1145/3774904.3792346.

[3] **Yueqing Xuan**, Kacper Sokol, Mark Sanderson, Jeffrey Chan. "Evaluating and Addressing Fairness Across User Groups in Negative Sampling for Recommender Systems". Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM), 2025, https://doi.org/10.1145/3746252.3761263.

[4] **Yueqing Xuan**, Kacper Sokol, Mark Sanderson, Jeffrey Chan. "Perfect Counterfactuals in Imperfect Worlds: Modelling Noisy Implementation of Actions in Sequential Algorithmic Recourse". Machine Learning, 114, no.8 (2025): 187, https://doi.org/10.1007/s10994-025-06821-1.

[5] Kacper Sokol*, Edward Small*, **Yueqing Xuan***. "Navigating Explanatory Multiverse through Counterfactual Path Geometry". Machine Learning, 114, no.6 (2025): 1-33, https://doi.org/10.1007/s10994-025-06769-2. *Equal contribution.

[6] **Yueqing Xuan**, Kacper Sokol, Mark Sanderson, Jeffrey Chan. "Leveraging Complementary AI Explanations to Mitigate Misunderstanding in XAI". 2025 IEEE Swiss Conference on Data Science (SDS), 2025, https://doi.org/10.1109/SDS66131.2025.00034.

[7] **Yueqing Xuan**, Edward Small, Kacper Sokol, Danula Hettiachchi, Mark Sanderson. "Comprehension Is a Double-Edged Sword: Over-Interpreting Unspecified Information in Intelligible Machine Learning

Explanations". International Journal of Human–Computer Studies (IJHCS), 2024, 103376, ISSN 1071-5819, https://doi.org/10.1016/j.ijhcs.2024.103376.

Workshop Papers

[1] Edward Small*, **Yueqing Xuan***, Danula Hettiachchi*, Kacper Sokol*. "Helpful, Misleading or Confusing: How Humans Perceive Fundamental Building Blocks of Artificial Intelligence Explanations". ACM CHI Workshop on Human-Centered Explainable AI (HCXAI). In: arXiv preprint arXiv:2303.00934. 2023.

[2] Kacper Sokol*, Edward Small*, **Yueqing Xuan***. "Navigating Explanatory Multiverse through Counterfactual Path Geometry". ICML (International Conference on Machine Learning) Workshop on Counterfactuals in Minds and Machines. In: arXiv preprint arXiv:2306.02786. 2023.

Theses

[1] **Yueqing Xuan**. "Accounting for User Diversity in the Pursuit of Fair and Explainable Machine Learning". PhD thesis. School of Computing Technologies, RMIT University, 2026.

[2] **Yueqing Xuan**. "Multimodal Entity Linker". Master's thesis. School of Engineering and Information Technology, University of Melbourne, 2021.

Academic Experience

Postdoctoral Research Fellow, RMIT University, Melbourne Mar 2026 – Present

- Investigate how users misunderstand AI explanations and how conversational guidance can improve user comprehension.
- Design and develop an interactive explanation navigator that guides users towards appropriate AI explanations based on their information needs.
- Research supported by the Google Research Scholar Program (2025).

Tutor, RMIT University, Melbourne Feb 2024 – Nov 2025

- Tutored undergraduate and postgraduate Algorithms and Analysis courses across four semesters.
- Led weekly tutorials and workshops, explaining algorithmic concepts and guiding problem-solving activities using Python.
- Marked assignments and examinations and provided academic support through student consultations.

Research Assistant, University of Melbourne, Melbourne Nov 2021

- Research project: Humanoid robotic arm control through verbal commands (GitHub Link).
- Integrated the Porcupine wake-word engine and IBM Watson speech recognition into a robotic arm control framework, enabling real-time interaction through natural-language voice commands.

Professional Activities

Peer Review: CHI (2027), DMKD (2026), WSDM (2026), SIGIR (2023, 2025).

Leadership: Organiser, RMIT Recommender and Responsible AI (R2AI) Reading Group.

Visiting Research: University of Amsterdam (2023); USI Lugano (2025).

Funding: ADM+S Research Training Support Fund (2025); RACE Merit Allocation Scheme (2026); Google Cloud Research Credits program (2026).